DA20 Build a reliable analysis

Week 7, Show it | Session 20 of 24 | Python for Data Analysis

Today's question. Can a stranger follow my steps from the raw file to my numbers?

Objective. By the end: you can build the calculations for your project so that each one is checked by hand and by a second route, and a partner can rerun them.

> **WHERE THIS IS USED**
> A result is only useful if someone else can repeat it. Analysts log each step and check each number two ways. Who could rerun your analysis and get the same numbers?

Before this session: DA19 Project question and a safe plan. If you cannot explain the last example, use its error card with the instructor first.

Make

Work from a protected copy of the raw data. Write the cleaning log, including a log that says no change was needed if that is true. Build one calculation at a time. For each, compare with a small hand-worked example and with a second route (pandas against NumPy, or a merged total against an unmerged one). Ask a partner to rerun one calculation from your notebook. Write down the first discrepancy they find and fix it.

Save your work as DA20. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
A calculation changes when the notebook is restarted	It depends on a value from a cell that no longer runs first.	Restart and run all. Move the first failing cell's dependency above it.
The partner gets a different answer	A step uses a file path or setting that only works on your laptop.	Use paths that sit beside the notebook and state them in the first cell.

Exit ticket

Every number in my notebook has a check that someone else can rerun.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.